

Alexei Ledenev

Principal Software Engineer — Agentic AI, Cloud Platforms & Kubernetes

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Summary

Principal engineer and cloud platform architect with **25+ years** building infrastructure, SaaS, and developer tools. Current focus: **agentic AI for cloud operations** — multi-agent systems, MCP tooling, safety guardrails, and human-approved execution across AWS, GCP, and Azure. Open-source maintainer of **Pumba** (3K+ stars), Kubernetes/AWS MCP servers (390+ stars combined), and **ccgram** (165+ stars); patent holder, speaker, and hands-on technical leader.

Selected Agentic AI & Platform Projects

- **Reflex** (DoiT, internal strategic initiative) — Leading architecture for an enterprise multi-agent cloud-operations platform with safety guardrails, approval gates, and controlled execution across cloud environments. *Implementation details proprietary.*
- **k8s-mcp-server** (Python, **210★**) — MCP server that lets AI assistants securely run `kubectl`, `helm`, `istioctl`, and `argocd` from a hardened containerized environment.
- **cloud-mcp-server / AWS CLI MCP** (Python, **182★**) — MCP server exposing AWS CLI workflows to Claude, Cursor, and other MCP clients with scoped credentials, container isolation, streamable HTTP transport, and MCP-native errors.
- **ccgram** (Python, **165★**) — Telegram ↔ tmux control plane for AI coding agents including Claude Code, Codex CLI, and Gemini CLI; supports remote monitoring, prompts, and parallel sessions.
- **cc-thingz** — Plugin suite for Claude Code, Codex, Pi, and Gemini; packages skills, agents, hooks, and cross-target build tooling for agentic software development.

Professional Experience

Principal Software Engineer — DoiT International | April 2025 – Present

- Leading **Reflex**, a strategic autonomous AI initiative for cloud operations, from architecture through guarded execution and human-in-the-loop workflows.
- Architecting migration of the **DoiT Cloud Intelligence (DCI)** platform from GCP to AWS/EKS for a multi-tenant enterprise FinOps SaaS product.
- Redesigning the analytics pipeline from **BigQuery to ClickHouse**; building the Go sync service for a 6-shard × 2-replica ClickHouse cluster.
- Driving BigQuery cost and performance optimization across the platform, reducing annual spend by **\$100K+**.

Staff Cloud Architect — DoiT International | September 2019 – April 2025

- Built and led a distributed team of senior Forward Deployed Engineers across AWS, GCP, Azure, AI/ML, containers, analytics, infrastructure, and automation.
- Owned architecture reviews, technical strategy, and best-practice governance for complex customer and platform engagements.
- Designed and shipped AI products integrated with DoiT Cloud Intelligence: **Zenrouter** for AI-driven ticket routing and an **LLM + OCR invoice-processing workflow** for AWS invoices.

- Drove cloud-native and FinOps adoption across customer engagements; delivered certified AWS/GCP training and spoke at cloud, Kubernetes, DevOps, and agentic-AI events.

Solutions Architect — Amazon Web Services | July 2018 – September 2019

- Advised enterprise customers on Well-Architected solutions across security, cost, performance, reliability, and operations; ran workshops, immersion days, and reference-architecture sessions.
- Specialized in Kubernetes and container platforms as a member of the **Containers Technical Field Community**.

Chief of Research — Codefresh | March 2017 – July 2018

- Led R&D on Kubernetes, Docker, and CI/CD for a SaaS microservices platform; maintained open-source projects and grew the Containers 101 TLV meetup community.

Chief Software Architect — Hewlett-Packard Enterprise, ADM Business Unit | 2013 – March 2017

- Principal architect for the Application Delivery Management business unit; led and mentored product architects across multiple enterprise products.
- Led innovation research and was awarded **9 patents** in test automation, including Mirror Testing and AI-driven Visual Object Recognition for UI automation.
- Drove migration of legacy enterprise products to SaaS, led acquisition due diligence, and supported the organization-wide Agile transition that reduced time-to-market by ~30%.

Earlier Roles | Software Engineer → Team Lead → R&D Manager → Architect | 1997–2013

- Hewlett-Packard, Mercury Interactive, Mer Technologies, Myriad Interactive. Progression through hands-on engineering, technical leadership, R&D management, and architecture roles.

Open Source & Community

- [Pumba](#) (Go, **3K+★**) — Chaos engineering for containers: network emulation, fault injection, and stress testing across Docker, Podman, and containerd.
- **Cloud-native OSS:** [KubeIP](#) (**447★**) for static Kubernetes node IPs; [Secrets-Init](#) (**172★**) for AWS/GCP secret injection; [Spotinfo](#) (**161★**) for AWS Spot inventory and pricing.
- **Technical writing and speaking** — Medium/ITNEXT articles and conference talks on AI coding agents, Kubernetes operations, cloud security, chaos engineering, and cloud-native architecture.

Skills

- **AI / Agentic Systems:** LLM application architecture, Model Context Protocol (MCP), multi-agent systems, safety guardrails, evaluation, LangChain, AutoGen, prompt and instruction design
- **Cloud / Platform:** AWS, GCP, Azure, Kubernetes, EKS, GKE, Docker, Cloud Run, BigQuery, ClickHouse, FinOps analytics
- **Engineering:** Go, Python, TypeScript, Shell, SaaS platforms, distributed systems, microservices, event-driven systems, platform engineering, modular architecture
- **DevOps / Leadership:** Terraform, GitHub Actions, CI/CD, IAM, secure sandboxed execution, distributed teams, hiring, mentoring, architecture governance, customer advisory

Education, Certifications & Patents

- **Education:** M.Sc. Computer Science, Tel Aviv University (2006–2007); B.Sc. Computer Science & Statistics, Tel Aviv University (1992–1996)
- **Certifications:** GCP Professional Cloud Architect & Authorized Trainer · AWS Solutions Architect – Professional & Authorized Instructor · CKA
- **Patents:** 9 granted patents in test automation, UI testing, protocol handling, and AI-driven automation.